

Adoption Is Not Security: Measuring End-to-End BGP Route Protection Across 80,000 Autonomous Systems

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Abstract — The Border Gateway Protocol (BGP) is the standard method of routing requests through the internet, but is vulnerable to prefix hijacking, where requests to a particular address may be routed toward the wrong destination. To mitigate this risk, the Internet Engineering Task Force (IETF) introduced the Resource Public Key Infrastructure (RPKI) in 2011 [1]. However, a significant portion of autonomous systems (ASes) have not adopted RPKI’s route origin authentication (ROA) or validation (ROV) [2], [3]. In this work, we evaluate the security of routes between ASes by using public datasets to construct a graph of AS routes, and traversing this graph using a bidirectional breadth-first search that satisfies the Gao-Rexford model. We searched for valid paths between two sets of AS pairs and categorized these paths based on ROV adoption of all ASes in the path, and the destination AS’s ROA adoption. We find that 56.8% of routes between stub ASes and major transit providers are secure, but this drops to 14.7% for randomly sampled AS pairs, revealing that RPKI deployment is heavily concentrated among large networks and that aggregate adoption statistics substantially overstate the Internet’s actual routing security.

1 Introduction

The Border Gateway Protocol (BGP) determines efficient ways to route data across the network of autonomous systems that makes up the Internet, and has been in use since 1994. However, BGP is suscepti-

ble to prefix hijacking attacks, in which an attacker uses IP addresses belonging to another AS to reroute traffic from that AS to itself [3]. Fortunately, Resource Public Key Infrastructure (RPKI) mitigates this vulnerability by allowing IP prefix owners to publish Route Origin Authorization (ROA) for specific ASes. Routers then utilize Route Origin Validation (ROV) to determine whether or not a specific route to an IP prefix is valid. However, many organizations have yet to implement RPKI due to bureaucratic difficulties in acquiring ROAs and fears that ROV adoption could drop legitimate, misconfigured traffic [4].

In this paper, we measure BGP route security by evaluating if paths between autonomous systems are actually protected by RPKI and detail the construction of a directed graph modeling the Internet’s AS topology. We then annotate this topology with real-world RPKI data, combining ROA coverage statistics with ROV enforcement scores from the ROVista project. Using this annotated graph, we analyze both large transit nodes and randomly selected AS pairs to quantify the vulnerability of network routes to prefix hijacking, demonstrating the differences in security between high-traffic core networks and edge nodes.

2 Background and Related Works

2.1 RPKI Deployment

Although development of the Resource Public Key Infrastructure (RPKI) started in 2008 and was first deployed in 2011, widespread adoption has been frus-

tratingly slow. Studies measuring RPKI implementation reveal that the vast majority of ASes do not use ROV, and the few that do only enforce it partially [4], [5].

We build on the work of Gilad et al. (2017) [5], who initially measured ROV enforcement in the Internet. They found that at least 78% of the top 100 ISPs did not enforce ROV at the time of the study. Furthermore, they demonstrated that major ISPs must almost unilaterally enforce ROV to get meaningful security benefits from RPKI, suggesting that a significant portion of routed traffic remained insecure. We expand upon their methodology to model actual cases of inter-AS path security, providing a graph-level perspective beyond individual AS enforcement statistics.

2.2 Valley-Free Routing

Gao and Rexford (2001) [6] devised a set of guidelines that ASes can implement independently to prevent conflicts resulting in route instability. Among these guidelines is the valley-free constraint, which means that a route cannot use a “downhill” edge and then use an “uphill” edge. This constraint prevents loops that can cause routing failures, ensuring that any paths found that meet the constraint are economically viable. We use this idea of a “valley-free” path, along with terminology from the paper such as “provider”, “peer”, and “customer” nodes to classify secure routes.

3 Methodology

3.1 Datasets

We use the ROV measurement framework, ROVista [2], created by Li et al. to classify whether an AS enforces ROV. The ROVista project provides ROV enforcement ratios for each AS inferred from control-plane observations. The dataset contains approximately 32,500 AS entries, each with a `ratio` field ranging from 0.0 (no observed ROV enforcement) to 1.0 (full enforcement). This covers roughly 26% of the ASes in our graph; the remaining 74% have no ROVista score. In our experiment, we use ROVista to determine whether or not a particular routing between ASes passes through secure or insecure nodes.

We use the CAIDA AS Relationships dataset, Serial-2 format, dated July 1, 2026, as the main source of our edge data [7]. This dataset provides inferred business

relationships between AS pairs. Each record is a pipe-delimited line of the form `AS1|AS2|rel_type|source`, where `rel_type` is `-1` (AS1 is a provider of AS2) or `0` (AS1 and AS2 are peers). This is our sole source of edge data. We chose CAIDA over ipverse connectivity data after discovering inconsistencies between the two datasets in customer-provider relationship classification; using a single authoritative source for edges avoids ambiguity.

The `ipverse.as.json` dataset provides per-AS metadata including a network role (transit, stub), and a connectivity reach score representing the AS’s customer cone size [8]. We use this data exclusively for optional node-level metadata enrichment and do not extract edge relationships from it.

We perform RPKI validation and generate a snapshot of the global Validated ROA Payload (VRP) set by running Routinator, a Relying Party software for RPKI [9]. We export its output in JSON format `authorisedroutes.json`. Each entry contains an origin ASN, an IP prefix, and a maximum prefix length. This provides the ground truth for which ASes have published ROAs and which prefixes they cover.

Our objective is to combine AS topology, ROA coverage, and ROV enforcement data into a single annotated graph, then apply a valley-free reachability algorithm that evaluates path-level security for sampled source-destination pairs. This produces a measurement that is more granular (per-pair security status) than separate, aggregate measurements of RPKI deployment.

3.2 Graph Construction and Annotation

To model the various relationships between ASes, we built a NetworkX directed graph (DiGraph) from the CAIDA dataset. For each provider-customer relationship (`rel_type = -1`), we add two directed edges: a `p2c` edge from provider to customer and a `c2p` edge from customer to provider. For each peering relationship (`rel_type = 0`), we add `p2p` edges in both directions. This bidirectional representation encodes the information needed by our reachability algorithm: when sitting at node A and examining outgoing edge `A→B`, the edge type indicates the direction of traversal relative to the business relationship (`c2p` = going uphill, `p2c` = going downhill, `p2p` = crossing a peer link).

All nodes are initialized with default security attributes: `has_roa = False`, `enforces_rov = False`, `rov_unknown = False`.

For each entry in the `ipverse as-metadata` dataset [8] whose ASN corresponds to an existing node in the graph, we attach the following metadata attributes: `name`, `country`, `country_code`, `category`, `network_role`, and `cone_size` (from the connectivity reach score). ASes present in the CAIDA graph but absent from `ipverse` retain their default empty attributes. This does not modify the graph’s edge structure.

We parse the VRP snapshot and group entries by origin ASN. For each ASN present in our graph, we set `has_roa = True` and store the full list of (`prefix`, `maxLength`) pairs as a node attribute. We also construct a `prefix_to_origin` mapping from each prefix string to its origin ASN; this mapping is used by the reachability algorithm to determine whether a destination’s address space is covered.

At the node level, `has_roa = True` means the AS has published at least one ROA for at least one of its prefixes. We do not distinguish between ASes with partial versus complete prefix coverage, as this would require mapping each AS’s announced prefix set against its ROA set — a measurement that requires additional data (e.g., a RIB snapshot) beyond the scope of this paper.

We load the ROVista dataset and apply a threshold-based classification. For each AS with a ROVista score ratio:

- If `ratio >= threshold`: `enforces_rov = True`, `rov_score = ratio`
- If `ratio < threshold`: `enforces_rov = False`, `rov_score = ratio`

For ASes not present in the ROVista dataset: `enforces_rov = False`, `rov_unknown = True`.

The default threshold is 0.5. We chose this value as a conservative midpoint; an AS with a ROVista score of 0.5 is filtering roughly half of RPKI-invalid routes, which provides partial but incomplete protection.

The 74% of ASes with `rov_unknown = True` constitute the primary source of uncertainty in our analysis. To keep a conservative estimate, unknown nodes are defaulted to `enforces_rov = False`.

3.3 Reachability Algorithm

Our reachability algorithm performs a bidirectional breadth-first search from a source AS to a destination AS, constrained by the Gao-Rexford valley-free property [6]. The search maintains a three-state machine:

- **UP**: The path is ascending (customer→provider). Valid transitions: `c2p` → UP, `p2p` → PEER, `p2c` → DOWN.
- **PEER**: The path has crossed a peer link. Valid transitions: `p2c` → DOWN only.
- **DOWN**: The path is descending (provider→customer). Valid transitions: `p2c` → DOWN only.

The search begins at the source node in the UP state. At each step, we examine outgoing edges from the current node and follow only those edges whose type is a valid transition from the current state. We track visited (`node`, `state`) pairs to prune redundant exploration, but exempt the destination node from the visited check so that it can be reached via multiple states (yielding distinct paths). We limit search depth to 10 hops, as most AS paths rarely exceed this length.

For each search run for a pair of ASes, we evaluate three conditions:

1. **Valley-free**: The path satisfies the Gao-Rexford constraints (guaranteed by construction).
2. **Destination has ROA**: The destination AS has `has_roa = True`.
3. **All intermediates enforce ROV**: Every AS on the path except the source and destination has `enforces_rov = True`.

A path is classified as **secure** if and only if all three conditions hold. The source AS is excluded from the ROV requirement because it is the party initiating the route lookup; it does not need to filter its own announcements. The destination AS is excluded from the ROV requirement because its role is to have published ROAs (proving its prefix ownership), not to filter incoming routes on behalf of the source.

3.4 Sampling Strategy

To produce a representative set of source-destination pairs, we sample from two populations:

Destinations. We select the top N ASes by customer cone size (default $N = 100$). These are the major transit providers and large networks whose reachability

is most consequential — they carry a disproportionate share of global traffic and appear as intermediate hops on a large fraction of paths.

Sources. We randomly sample M stub ASes (default $M = 200$). Stub ASes are identified by their `ipverse network_role` attribute or, as a fallback, by having zero provider-to-customer edges in the graph. These represent end-user networks, enterprises, and content providers — the entities that initiate traffic.

We construct pairs as follows: every sampled stub is paired with every top- N AS, producing up to $N \times M$ stub-to-transit pairs (simulating end users reaching major networks). We also include bidirectional top- $N \times$ top- N pairs (transit backbone reachability). If the total exceeds a configurable cap (default 5,000), we subsample uniformly at random with a fixed seed for reproducibility.

Uniform Random Baseline. In addition to the structured stub-to-transit pairs, we generate a baseline evaluation using uniform random sampling via a separate Python sampling script. We filter `as.json` for all active ASNs containing a valid `lastAnnounced` timestamp and draw 10,000 unique source-destination pairs uniformly at random. This baseline complements our stratified sampling by covering arbitrary leaf-to-leaf and general-transit routing scenarios across the full AS population.

4 Results

After running our algorithm on the pairs described in the previous section, we obtained the following results.

	Stub-to-Transit	Random
Total Pairs Tested	5,000	10,000
Valley-Free Reachable	4,639 (92.8%)	7,856 (78.6%)
Secure Paths	2,838 (56.8%)	1,470 (14.7%)
Insecure: No Dest. ROA	412 (8.2%)	3,219 (32.2%)
Insecure: Transit ROV Gap	1,389 (27.8%)	3,167 (31.7%)
No Valley-Free Path	361 (7.2%)	2,144 (21.4%)

As shown in the table above, the two sampling strategies reveal a stark contrast in BGP route security

between an average pair of ASes and between larger, more established ASes. We see that only 14.7% of routes between a random pair of ASes are secure, while 56.8% of those between larger transit nodes are secured, implying that overall adoption of the RPKI framework is concentrated within larger ASes, as one might expect.

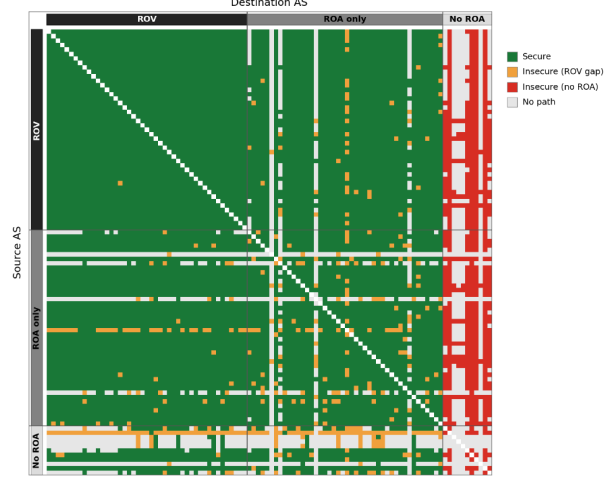


Figure 1: Pairwise path security among the top 100 ASes by customer cone size. each cell shows the security status of the valley-free path from source (row) to destination (column), with ASes grouped by RPKI posture: ROV-enforcing (45), ROA-only (44), and no ROA (11). the rov-enforcing block is near-uniformly secure, while the rightmost columns show that any path to a destination without ROA coverage is inherently insecure regardless of source posture. The scattered white bands in the ROA-only block reflect ASes with limited valley-free reachability at the search depth used.

It should be noted that this stark difference largely comes from the destination nodes of the randomly selected pairs not having ROA, with only 49.6% of random nodes having them, while 85.5% of stub-to-transit have ROA. In this instance, the theory that higher-traffic nodes are more likely to adopt RPKI still holds, as regardless of whether or not a destination node enforces ROA, most of the traffic throughout the network will likely route through the higher-traffic nodes, which gives us a similar ratio of routes having ROV enforcement.

5 Discussion

Our results show that the fraction of securely routable AS pairs varies dramatically depending on which parts of the internet are examined: 56.8% for traf-

fic toward major transit providers, but only 14.7% across the general AS population. However, this conclusion is based upon partially incomplete data. Any future developments in mapping or simulating the impacts of RPKI coverage could benefit from a larger set of ROV enforcement measurements and from an improved reachability algorithm based on more concrete routing data.

In the case of ROV enforcement data, RoVista’s publicly available dataset only covers roughly a quarter of the ASes listed in the ipverse dataset, as the scores for nodes with extremely low traffic were dropped [2]. Expanding data of ROV enforcement to cover more nodes could be beneficial, although we believe it would not influence the scores significantly, as we have shown that lower-traffic ASes tend to route traffic towards the same core high-traffic nodes, so the ROV enforcement rates would likely only see a minor decrease.

Our findings also illustrate the progress that ROV has made over the past decade. Gilad et al. (2017) [5] found that at least 78 of the top 100 ISPs did not enforce ROV at all, but the measurements show that 89 of the top 100 ISPs now have ROAs. While this suggests significant growth, it should be interpreted cautiously, as ROA registration is insufficient for ROV enforcement.

Other researchers could also improve the accuracy of the reachability algorithm. While our current algorithm does utilize known relationships between connected ASes, the actual routes produced are simply possible routes that fit the Gao-Rexford model, which may not exactly align with the routes used in practice. Having more granular knowledge of the factors used in routing, or implementing routing table data, could yield more accurate results.

6 Conclusion

Regardless of any limitations of our current work, our current conclusions are consistent with other research findings in that the present rate of RPKI adoption does not allow us to confidently state that routes over the internet are secured. For a more secure internet, ASes with more customers and connections should continue to adopt ROV, as most of the routes on the internet pass through them. At the same time, IP owners should issue more ROAs for their ASes, even if they do not enforce ROVs themselves, as lacking an ROA leaves the service vulnerable to BGP hijacks or simple misconfigurations, even if the rest of the internet adopts ROV.

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